

Engineering Program Management PoC

CynLr Robotics

Engineering Program Planning,
Dependency Management,
Resource Planning and KPI Framework

Prepared by

Samay Thakur

Electronics & Instrumentation Engineering
M.S. Ramaiah Institute of Technology, Bengaluru

Tools Used

Asana — Google Sheets / Microsoft Excel — GitHub

September 2026

1 Executive Summary

This document presents an Engineering Program Management plan developed for a CynLr robotics Proof of Concept (PoC). The objective of the exercise was to convert a technically complex robotics problem into a structured and executable program plan.

The program plan was developed in Asana and supported by a resource and effort estimation workbook. The planning approach covers phased technical execution, task and intra-project dependencies, external dependencies, interdisciplinary coordination, risks and constraints, resource planning, KPIs, lead and lag indicators, testing, and final delivery.

The program was structured around two primary technical phases:

- Phase 1 – Track Moving Objects
- Phase 2 – Track and Grasp

Additional workstreams were created for Hardware and Integration, Testing and Validation, External Dependencies, Risks and Decisions, Program Management, Resource and Effort Planning, and Final Review.

The resulting plan provides a structured view of how robotics, computer vision/ML, software, mechanical, electronics, procurement, and external stakeholders can coordinate toward a common PoC outcome.

2 Project Overview

The CynLr PoC involves demonstrating robotic tracking and grasping capabilities for moving objects. From an Engineering Program Management perspective, the challenge is not limited to the technical implementation. It also requires coordination between multiple engineering disciplines, hardware availability, software dependencies, external vendors, procurement, testing, and final delivery.

The program plan therefore focuses on:

- Breaking the PoC into manageable phases and activities.
- Establishing dependencies between activities.
- Identifying external dependencies and potential blockers.
- Mapping interdisciplinary engineering activities.
- Identifying required resources and estimated skill hours.
- Defining risks, constraints, and mitigation actions.
- Establishing KPIs and success indicators.
- Defining testing and final delivery criteria.

3 Program Objectives

The primary objectives of the program plan are:

1. Develop a structured Engineering Program Management plan for the robotics PoC.
2. Break the technical objective into executable phases, activities, and subtasks.
3. Map task, intra-project, and external dependencies.

4. Identify interdisciplinary interfaces and coordination requirements.
5. Estimate resources and skill hours required for major activities.
6. Identify project risks, constraints, and potential bottlenecks.
7. Establish measurable KPIs and lead/lag indicators.
8. Define testing, review, and final delivery mechanisms.

4 Program Structure

The project was divided into the following major workstreams:

Workstream	Purpose
Phase 1 – Track Moving Objects	Demonstrate tracking of moving objects using robot, camera, and conveyor systems.
Phase 2 – Track & Grasp	Extend tracking capability toward grasping using robot freedom and RTU integration.
Hardware & Integration	Integrate and validate robot, conveyor, camera, RTU, and related hardware.
Testing & Validation	Define test cases, execute end-to-end tests, and document performance.
External Dependencies	Track dependencies involving manufacturing partners, procurement, software libraries, and external support.
Risks & Decisions	Identify risks, constraints, mitigation actions, and program decisions.
Program Management	Track milestones, coordinate teams, manage blockers, and maintain program-level visibility.
Resource & Effort Planning	Identify resources and estimate skill hours for program activities.
Final Review	Review overall PoC results, document findings, and prepare final demonstration/handover.

Table 1: Major program workstreams

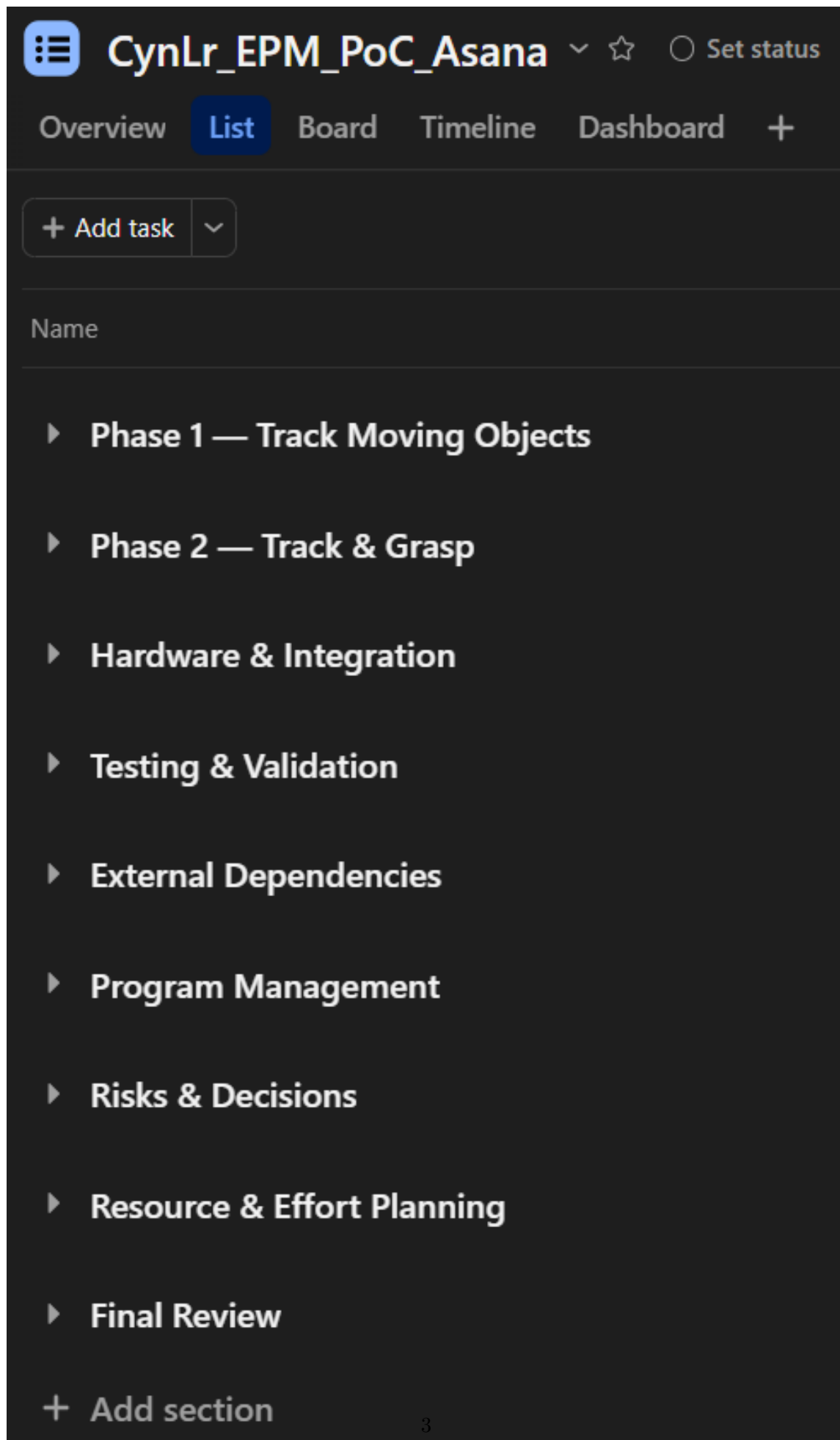


Figure 1: Asana project structure and major workstreams.

5 Phase 1 – Track Moving Objects

Phase 1 focuses on demonstrating tracking of moving objects using a robot, camera, and conveyor setup.

The planned sequence includes:

1. Define Phase 1 experiment.
2. Verify robot and conveyor availability.
3. Prepare camera and sensing setup.
4. Calibrate camera and coordinate system.
5. Develop object-tracking capability.
6. Integrate tracking output with the robot.
7. Execute tracking experiments.
8. Analyze experiment results.
9. Conduct Phase 1 review.

The sequencing establishes technical readiness before experimentation and creates a review gate before progressing to the next phase.

Name	Due date	Assignee	Phase	
▼ Phase 1 — Track Moving Objects				
▶ ☑ Define Phase 1 Experiment 5 📅	📅	👤	Phase 1 — Track M.	
▶ ⌘ Verify Robot and Conveyor Availability 5 📅	📅	👤	Phase 2 — Track &	
▶ ⌘ Prepare Camera and Sensing Setup 5 📅	📅	👤	Hardware & Integr.	
▶ ⌘ Calibrate Camera and Coordinate System 4 📅	📅	👤	Testing & Validation	
▶ ⌘ Develop Object Tracking Capability 5 📅	📅	👤	External Dependen	
▶ ⌘ Integrate Tracking Output with Robot 5 📅	📅	👤	Program Manage...	
▶ ⌘ Execute Tracking Experiments 5 📅	📅	👤	Risks & Decisions	
▶ ⌘ Analyze Experiment Results 5 📅	📅	👤	Final Review	
▶ ⌘ Conduct Phase 1 Review 4 📅	📅	👤		
Add task				

Figure 2: Phase 1 activities.

6 Phase 2 – Track and Grasp

Phase 2 builds on the tracking capability and introduces grasping and RTU integration.

The planned activities include:

1. Define Phase 2 experiment.
2. Confirm RTU and robot availability.
3. Prepare RTU integration setup.
4. Integrate robot and RTU.

5. Develop track-and-grasp capability.
6. Integrate gripper or end-of-arm tooling.
7. Execute grasping experiments.
8. Evaluate multi-robot coordination feasibility.
9. Conduct Phase 2 review.

Phase 2 is dependent on sufficient Phase 1 progress and the availability of the required hardware and integration infrastructure.

Name	Due date	Assignee	Phase
Phase 2 — Track & Grasp			
Define Phase 2 Experiment 5			Phase 1 — Track M.
Confirm RTU and Robot Availability 5			Phase 2 — Track & Grasp
Prepare RTU Integration Setup 5			Hardware & Integr.
Integrate Robot and RTU 3			Testing & Validation
Develop Track-and-Grasp Capability 3			External Depend.
Integrate Gripper / End-of-Arm Tooling 3			Program Manage...
Execute Grasping Experiments 3			Risks & Decisions
Evaluate Multi-Robot Coordination Feasibility 3			Final Review
Conduct Phase 2 Review 3			
Add task			

Figure 3: Phase 2 activities.

7 Dependency Management

Dependency management was incorporated at three levels:

7.1 Intra-Project Dependencies

These represent relationships between activities within the program. Examples include:

- Camera setup before camera calibration.
- Calibration before object-tracking development.
- Tracking development before robot integration.
- Phase 1 review before progression into Phase 2 execution.
- Hardware integration before end-to-end testing.
- Test-case definition before test execution.
- Test execution before performance analysis.

7.2 External Dependencies

The program also includes dependencies outside the immediate engineering team:

- Hardware manufacturing partners.

- Procurement of imported components.
- Robot kinematic library support.
- FANUC support.
- Hardware availability and delivery.

Name	Due date	Assignee	Phase
▼ External Dependencies			
🕒 Identify Required External Resources	📅	👤	Phase 1 — Track M.
🕒 Track Hardware, Software, and Access Dependencies	📅	👤	Phase 2 — Track &
🕒 Resolve Dependency Blockers	📅	👤	Hardware & Integr.
Add task			

Figure 4: Dependency relationships represented in the Asana program plan.

8 Interdisciplinary Coordination

The PoC requires coordination across several engineering disciplines.

Discipline	Program Contribution
Robotics	Robot availability, integration, motion and grasping activities.
Computer Vision / ML	Object tracking and vision-related development.
Software Engineering	Tracking integration, communication and software interfaces.
Mechanical Engineering	Gripper and end-of-arm tooling integration.
Electronics Engineering	Hardware communication and system integration.
Procurement / Finance	Component procurement and external dependency coordination.
Program Management	Planning, dependency management, risk tracking and reviews.
External Vendors	Manufacturing and external technical support.

Table 2: Interdisciplinary engineering interfaces

The program structure is designed to make handoffs and dependencies visible rather than treating each discipline as an isolated workstream.

9 Resource and Effort Planning

A separate resource and effort estimation workbook was prepared to provide a tabulated walk-through of activities and subtasks.

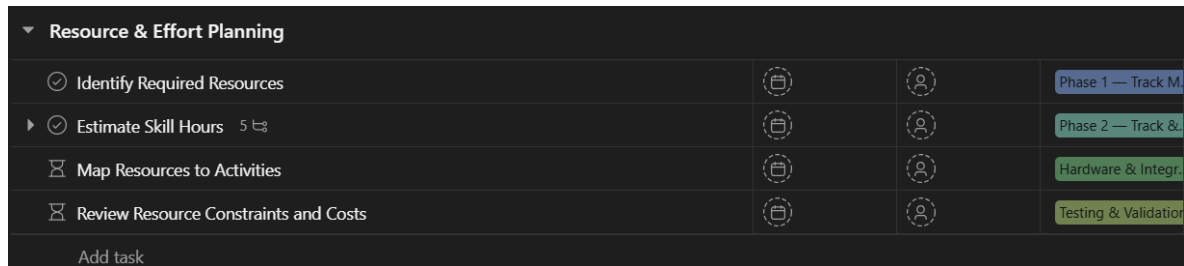
The workbook maps:

- Activity
- Description
- Sub-Activity
- Resource Category
- Estimated Effort in Hours

- Location

The preliminary activity-level estimates provide a program-level view of engineering effort and help identify resource-intensive areas.

The current planning model contains a preliminary estimated effort of approximately **604 person-hours**. These values represent planning estimates rather than measured execution effort.



▼ Resource & Effort Planning			
🕒 Identify Required Resources	📅	👤	Phase 1 — Track M.
▶ 🕒 Estimate Skill Hours 5 📅	📅	👤	Phase 2 — Track &.
🕒 Map Resources to Activities	📅	👤	Hardware & Integr.
🕒 Review Resource Constraints and Costs	📅	👤	Testing & Validation
Add task			

Figure 5: Resource and effort estimation table.

10 Risks, Bottlenecks and Constraints

Key risks and constraints considered in the program include:

- Limited simulation or virtual validation capability.
- Dependence on external hardware manufacturing partners.
- Procurement lead times for imported components.
- Availability and compatibility of robot kinematic libraries.
- Dependence on FANUC technical support.
- Hardware and software integration issues.
- Tracking and grasping reliability during experimental validation.

The program plan addresses these risks through dependency tracking, readiness checks, escalation mechanisms, phase reviews, and contingency planning.

11 KPI and Success Measurement Framework

KPIs were defined to evaluate both program execution and technical outcomes.

11.1 External Dependencies

- Number of unresolved external blockers.
- Average dependency resolution time.
- Percentage of critical dependencies resolved before the required milestone.

11.2 Research and Development

- Research milestones completed.
- Time taken to validate a technical approach.

- Development tasks completed.
- Integration defects identified and resolved.
- Percentage of planned development milestones completed.

11.3 Infrastructure

- Robot, camera, conveyor, RTU, gripper and software readiness.
- Number of infrastructure blockers.
- Percentage of required setup activities completed.

11.4 Testing and Validation

- Test-case completion rate.
- Object-tracking success rate.
- Grasping success rate.
- End-to-end test success rate.
- Number of critical defects.

11.5 Delivery

- Milestone completion rate.
- Deliverables completed.
- Number of unresolved critical issues at handover.
- Final PoC acceptance status.

Name	Due date	Assignee	Phase	
▼ Program Management				
✓ Track Project Progress and Milestones			Phase 1 — Track M	
⌘ Coordinate Team Activities and Reviews			Phase 2 — Track &	
⌘ Maintain Project Documentation and Status Upd.			Hardware & Integr.	
▶ ✓ Define KPIs and Progress-Balancing Scheme 4			Testing & Validation	
Add task				
				Subtasks 0/4 +
				✓ Define KPIs for Technical Performance
				✓ Define KPIs for Research, Development, and Infrastructure
				✓ Define Lead and Lag Indicators
				✓ Define Progress-Balancing and Escalation Mechanism
				Add subtask

Figure 6: KPI and success-metrics framework.

12 Lead and Lag Indicators

12.1 Lead Indicators

Lead indicators provide early visibility into whether the program is moving toward the intended outcome.

- Hardware and software readiness.
- Completion of calibration.
- Test cases defined.

- Dependencies assigned to owners.
- Blockers resolved within agreed timelines.
- Milestones completed as planned.

12.2 Lag Indicators

Lag indicators measure the resulting technical and delivery outcomes.

- Tracking performance.
- Grasping performance.
- End-to-end system reliability.
- Final PoC acceptance.
- Completion of agreed deliverables.

13 Progress-Balancing Mechanism

The program uses a progress-balancing approach rather than evaluating progress only through task completion.

The mechanism includes:

- Recognition for completing milestones and meeting agreed quality criteria.
- Recognition for resolving critical dependencies proactively.
- Escalation of delayed or blocked activities during program reviews.
- Reassessment of estimates when activities consistently exceed planned effort.
- Resource reallocation when bottlenecks affect critical milestones.
- Evaluation of quality and integration readiness in addition to task completion.

This approach provides a mechanism for balancing schedule, quality, resources, dependencies and technical outcomes.

14 Testing and Validation

Testing was structured as a dedicated program workstream.

The testing sequence includes:

1. Define test cases and success criteria.
2. Execute end-to-end tracking and grasping tests.
3. Measure performance and document results.
4. Review technical outcomes and unresolved issues.

Testing is linked to hardware and infrastructure readiness so that end-to-end validation is performed only after the required integration activities are complete.

15 Program Management and Review Gates

Program-level activities provide coordination and visibility across technical workstreams.

These activities include:

- Tracking project progress and milestones.
- Coordinating team activities and reviews.
- Resolving dependency blockers.
- Maintaining program-level documentation.
- Reviewing risks and decisions.
- Monitoring KPIs and success criteria.

Phase reviews act as decision points before progressing to subsequent stages.

16 Final Review and Delivery

The final review consolidates the results of Phase 1, Phase 2, hardware integration, and testing.

The final delivery sequence includes:

1. Review overall PoC results.
2. Document final findings and recommendations.
3. Prepare final demonstration and handover.

The final review considers technical outcomes, unresolved issues, dependencies, risks, and completion of agreed deliverables.

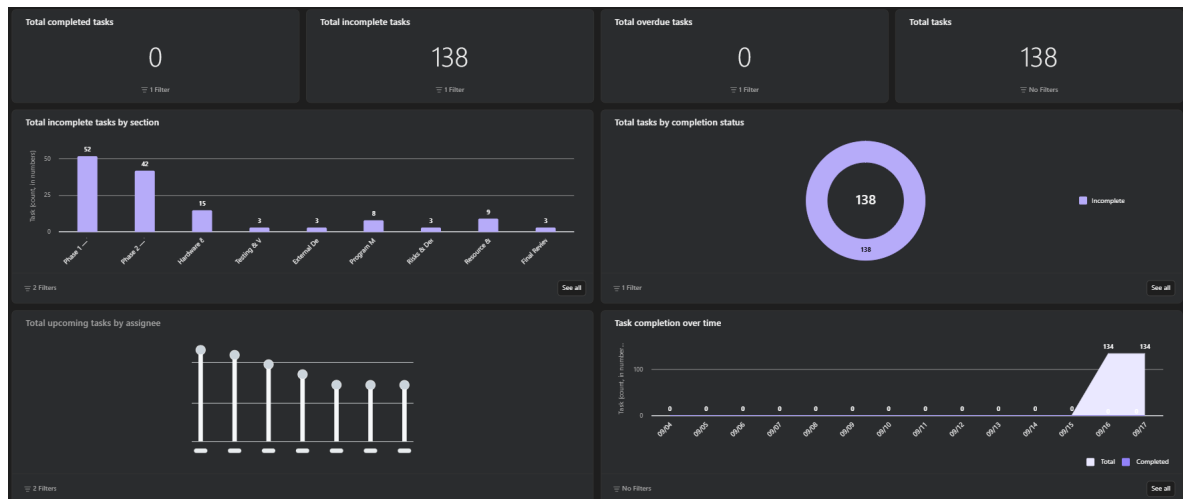


Figure 7: Asana dashboard providing program-level visibility.

17 Tools and Artifacts

The following tools were used for the program-planning exercise:

Tool	Purpose
Asana	Project structure, task management, subtasks, dependencies, program reviews and KPI planning.
Google Sheets / Microsoft Excel	Activity/subactivity walkthrough, resource categories, estimated skill hours and location mapping.
GitHub	Version-controlled documentation and project artifact organization.
$\text{\LaTeX}$ / Overleaf	Formal project documentation and final report preparation.

Table 3: Tools used

18 Conclusion

This Engineering Program Management PoC demonstrates how a multidisciplinary robotics initiative can be structured into an executable program plan.

The resulting framework connects technical phases with dependencies, external constraints, interdisciplinary resources, risks, testing, KPIs, progress indicators and final delivery. The combination of an Asana-based execution structure and a resource-effort estimation workbook provides both operational visibility and program-level planning.

The plan is intended as a structured PoC planning artifact rather than a claim of actual production execution. Effort values, performance metrics and success thresholds are therefore treated as planning criteria to be validated during actual execution.

A Project Artifacts

The following artifacts are included as part of the submission:

- Asana project link.
- Asana project CSV/XLSX export.
- Resource and effort estimation workbook.
- KPI and success metrics sheet.
- Project documentation.

B Asana Project Link

Asana:

<https://app.asana.com/1/1218515969146377/project/1218521240190043/list/1218523044337711>
https://docs.google.com/spreadsheets/d/1C9NnafF7m4unquux8jyWNJvI3ZUCDSK3/edit?usp=drive_link&ouid=117121494467764913352&rtpof=true&sd=true

C Resource and Effort Sheet

Google Sheets:

https://docs.google.com/spreadsheets/d/1cw95R0yf0Rgp7FZBRMEmlxW_rCNjbu8E/edit?usp=drive_link&ouid=117121494467764913352&rtpof=true&sd=true